THE UNIVERSAL QUOTIENT OF A COHERENT MODULE AWAY FROM THE POINTS OF A SPECIALIZATION-STABLE SUBSET, WITH DEPTH AT LEAST ONE THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The subsheaf of sections whose support is contained in a specialization-stable subset, and its description as the filtered colimit of the sections with support in the closed subsets contained in it. The author's section on supports of modules and its local description (source line 3325), used together with a lemma of the chapter on dualizing complexes. Identifies the kernel of the required quotient and makes the depth condition a condition on associated primes.
- Flat base change for such quotients, in the form recorded in the chapter on divisors. Invoked inside the counted proof at source line 3315-3316 as a cross-chapter pointer. Gives the compatibility of the construction with flat morphisms once uniqueness is known.

Source reference. Original reference divisors-lemma-bourbaki: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 3316–3316. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-bourbaki, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dualizing-lemma-divide-by-torsion: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1513–1513. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-divide-by-torsion, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's

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Lemma 0.1. *Let X be a Noetherian scheme. Let $T \subset X$ be a subset stable under specialization. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then there is a unique map $\mathcal{F} \to \mathcal{F}'$ of coherent $\mathcal{O}_X$-modules such that*

(1) *$\mathcal{F} \to \mathcal{F}'$ is surjective,*
(2) *$\mathcal{F}_x \to \mathcal{F}'_x$ is an isomorphism for $x \notin T$,*
(3) *$\text{depth}_{\mathcal{O}_{X,x}}(\mathcal{F}'_x) \geq 1$ for $x \in T$.*

If $f : Y \to X$ is a flat morphism with Y Noetherian, then $f^*\mathcal{F} \to f^*\mathcal{F}'$ is the corresponding quotient for $f^{-1}(T) \subset Y$ and $f^*\mathcal{F}$.

Proof. Condition (3) just means that $\text{Ass}(\mathcal{F}') \cap T = \emptyset$. Thus $\mathcal{F} \to \mathcal{F}'$ is the quotient of $\mathcal{F}$ by the subsheaf of sections whose support is contained in T. This proves uniqueness. The statement on pullbacks follows from Divisors, Lemma divisors-lemma-bourbaki and the uniqueness.

Existence of $\mathcal{F} \to \mathcal{F}'$. By the uniqueness it suffices to prove the existence and uniqueness locally on X; small detail omitted. Thus we may assume $X = \text{Spec}(A)$ is affine and $\mathcal{F}$ is the coherent module associated to the finite A-module M. Set $M' = M/H^0_T(M)$ with $H^0_T(M)$ as in Section section-supports. Then $M_\mathfrak{p} = M'_\mathfrak{p}$ for $\mathfrak{p} \notin T$ which proves (1). On the other hand, we have $H^0_T(M) = \text{colim}\, H^0_Z(M)$ where Z runs over the closed subsets of X contained in T. Thus by Dualizing Complexes, Lemmas dualizing-lemma-divide-by-torsion we have $H^0_T(M') = 0$, i.e., no associated prime of M' is in T. Therefore $\text{depth}(M'_\mathfrak{p}) \geq 1$ for $\mathfrak{p} \in T$. □